

OBLIGATORY TRIPLE SYSTEMS: AN ALTERNATIVE PROOF, LEAN FORMALISATION, AND FINITE CONSEQUENCES

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ABSTRACT. Eric Li's June 2026 preprint gave the first publicly posted complete classification of the finite triple systems that occur in every triple system of uncountable chromatic number. We give an alternative proof of that classification, a Lean formalisation, and further finite consequences. After isolated vertices are removed, obligatoriness is equivalent to linearity, incidence of every hyperedge-node of the Levi graph with a bridge, and evenness of every Berge cycle; equivalently, the systems are assembled from private-vertex expansions of finite bipartite graphs by disjoint unions and one-point amalgamations. The positive direction uses a probabilistic rainbow lemma and rooted abundance. The converse uses the one-apex sequence-lift and bridge-trace framework introduced by Li, reorganised as a base-fibre and support-incidence analysis. As consequences of the classification, we determine the exact possible triples consisting of the number of hyperedges, the number of vertices, and the number of connected components; derive the exact Levi cycle-rank spectrum; give a forbidden-Levi-subgraph formulation; and obtain balanced-hypergraph and colouring-polynomial consequences.

INTRODUCTION

The hosts in this problem are infinite, but the answer is decided by a small piece of finite geometry. A finite triple system is *obligatory* if it occurs in every triple system of uncountable chromatic number. Erdős asked for a characterization of these finite systems [1]; the question is catalogued as Erdős Problem 593 [2]. The answer has both a constructive form and an intrinsic form.

For a finite graph J , write J^+ for its private-vertex expansion. Let $\mathcal{B}$ be the smallest class containing J^+ for every finite bipartite graph J and every finite edgeless triple system, and closed under finite disjoint unions and one-point amalgamations. If F is a triple system, let F° denote the system obtained by deleting its isolated vertices.

Theorem A. *For every finite triple system F , the following are equivalent:*

- (1) F is obligatory;
- (2) $F \in \mathcal{B}$;
- (3) F° is linear, every hyperedge-node of $I(F^\circ)$ is incident with a bridge, and every Berge cycle of F° has even length.

Theorem A, including both the constructive and intrinsic formulations, was first proved by Li [3, Theorem 1.1]. Li also introduced the complete-rank one-apex sequence lift and exact bridge-trace theorem used in the negative half of the classification [3, Sections 3–4 and Theorem 4.6]. The present paper gives an alternative implementation: direct positive-atom and closure proofs, an explicit base-fibre and support-incidence form of the finite trace theorem, a complete Lean formalisation, and the finite consequences in Section 10. Dependencies on Li's proof are cited where they enter.

The graph analogue was proved by Erdős and Hajnal: the obligatory finite graphs are exactly the bipartite graphs [4, Corollary 5.6 and Theorem 7.4]. The classical nonlinearity obstruction for uniform hypergraphs is due to Erdős, Hajnal, and Rothschild [5, Theorem 2, p. 532]. Obligatory triple systems were studied further by Erdős et al. [6], Komjáth [7], Hajnal and Komjáth [8], and Komjáth [9]; the expansion theorem of Reiher [10, Theorem 1.2] supplies the positive atoms used below.

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Relation to Li's proof. Li's preprint has priority for Theorem A and for the one-apex and bridge-trace framework. The selected-incidence decomposition, quotient forest, and running-intersection assembly are also cited at their points of use [3, Sections 3–5]. The implementation here differs in its direct proof of the positive expansion atom, its rooted-abundance closure argument, its base-fibre/support-incidence organisation, its Lean development, and the additional finite deductions of Section 10.

The proof is best read as two stories. On the positive side, a probabilistic rainbow lemma provides the local injectivity needed to force complete bipartite expansions. A rooted-abundance lemma then proves closure under one-point amalgamation, including at singular uncountable cardinals. Deleting the bridges of the Levi graph identifies each remaining active component with the expansion of a finite bipartite graph; the quotient graph is a forest, and its running-intersection property gives the required amalgamation order.

On the negative side, we use Li's one-apex sequence-lift strategy [3, Sections 3–4]. The lift transfers uncountable chromatic number from a graph to a triple system while controlling every finite linear trace. We prove the required trace statement through expansion fibres and a support-incidence forest, so that the fibres are joined only at cut points. This prevents a bridge-free hyperedge-node or a Berge cycle from being assembled across different fibres. There are exactly three ways to fail the intrinsic test, and each has its own host: an Erdős–Rado linear host excludes nonlinearity, the lift of K_{ω_1} excludes a missing bridge, and a classical Erdős–Hajnal graph of uncountable chromatic number and prescribed odd girth excludes an odd Berge cycle.

All embeddings are injective and non-induced. The argument is carried out in ZFC. The imported results used as black boxes are the de Bruijn–Erdős compactness theorem [11, Theorem 1, pp. 371–373], the Erdős–Hajnal high-odd-girth theorem [4, Theorem 7.4, p. 76], and the pair case of the Erdős–Rado partition theorem [12, Theorem 4(i), formula (95), p. 471].

1. PRELIMINARIES

A *triple system* is a simple 3-uniform hypergraph $H = (V(H), E(H))$, so $E(H) \subseteq [V(H)]^3$. A colouring $c : V(H) \rightarrow \lambda$ is *proper* when no hyperedge is monochromatic, and $\chi(H)$ is the least cardinal λ admitting such a colouring. An embedding of a triple system F in H is an injective map $f : V(F) \rightarrow V(H)$ such that $f[e] \in E(H)$ for every $e \in E(F)$; embeddings are not required to be induced. A finite triple system F is *obligatory* if every triple system H with $\chi(H) > \aleph_0$ contains an embedding of F . A point is *isolated* if it belongs to no hyperedge, and F° denotes the result of deleting all isolated points.

All ordinary graphs in this paper are simple. A triple system is *linear* when any two distinct hyperedges meet in at most one point. Its *Levi graph* $I(F)$ is the bipartite graph with classes $V(F)$ and $E(F)$, where a point-node p is adjacent to a hyperedge-node e exactly when $p \in e$. A *bridge* is an edge of a graph whose deletion increases the number of connected components. A *Berge cycle of length $\ell \geq 2$* consists of distinct points $p_0, \dots, p_{\ell-1}$ and distinct hyperedges $e_0, \dots, e_{\ell-1}$ such that $p_i \in e_i \cap e_{i+1}$, with indices modulo ℓ . Equivalently, it is a simple cycle of length 2ℓ in $I(F)$.

For a finite simple graph J , its *private-vertex expansion* J^+ has vertex set

$$V(J) \dot{\cup} \{p_a : a \in E(J)\}$$

and hyperedges $\{u, v, p_{\{u,v\}}\}$ for $\{u, v\} \in E(J)$, where the points p_a are new and pairwise distinct. A *one-point amalgamation* of vertex-disjoint triple systems F_0, F_1 is obtained by choosing $x_i \in V(F_i)$, identifying x_0 with x_1 , and making no other identifications or new hyperedges. Finite disjoint unions and finite edgeless systems include the empty cases.

Finite reductions.

Lemma 1.1 (Isolated-vertex reduction). *Let F be a finite triple system and let F° be obtained by deleting all isolated vertices. Then*

$$F \text{ is obligatory} \iff F^\circ \text{ is obligatory,}$$

and

$$F \in \mathcal{B} \iff F^\circ \in \mathcal{B}.$$

Proof. If F° is obligatory and H has uncountable chromatic number, then H has infinitely many vertices. After embedding the finite system F° , map the isolated vertices of F injectively to unused host vertices. Containment is non-induced, so additional host edges are irrelevant. The converse follows because F° is a subhypergraph of F .

For membership in $\mathcal{B}$, the implication $F^\circ \in \mathcal{B} \Rightarrow F \in \mathcal{B}$ follows by taking the disjoint union with a finite edgeless system.

For the converse, use structural induction on a construction of F in $\mathcal{B}$. If $F = J^+$, delete the isolated graph vertices of J to obtain J_0 ; then $(J^+)^{\circ} = J_0^+ \in \mathcal{B}$. If F is edgeless, its reduction is the empty edgeless system. Reduction commutes with disjoint union.

Suppose finally that F is a one-point amalgamation of F_0 and F_1 , identifying $x_i \in V(F_i)$. By induction, $F_0^{\circ}, F_1^{\circ} \in \mathcal{B}$. If both selected vertices are nonisolated in their factors, then F° is the one-point amalgamation of the two reductions at those vertices. If exactly one selected vertex is nonisolated, the isolated side disappears at the identified point and F° is the disjoint union of the two reductions. The same description applies when both selected vertices are isolated, because the identified point is then deleted. Hence $F^\circ \in \mathcal{B}$ in every case. $\square$

Thus isolated vertices may be suppressed throughout the substantive argument and restored at the end.

Lemma 1.2 (Finite deletion). *If $\chi(H) > \aleph_0$ and $S \subseteq V(H)$ is finite, then*

$$\chi(H - S) > \aleph_0.$$

Proof. Otherwise a countable colouring of $H - S$, together with finitely many fresh colours for S , would colour H countably. $\square$

Colouring and closure tools.

Lemma 1.3 (Two elementary colouring facts). *We shall use the following two facts repeatedly.*

1. *If $E(H) = E_1 \cup \dots \cup E_r$ and every $(V(H), E_i)$ is countably chromatic, then H is countably chromatic.*
2. *If a graph G admits an orientation in which every vertex has outdegree at most $d < \omega$, then $\chi(G) \leq 2d + 1$.*

Proof. For the first assertion, take the product of the r countable colourings.

For the second, every finite subgraph has at most $d|V|$ edges and therefore a vertex of degree at most $2d$. It is $(2d + 1)$ -colourable by greedy deletion. The de Bruijn–Erdős compactness theorem [11, Theorem 1] then gives a $(2d + 1)$ -colouring of all of G . $\square$

Lemma 1.4 (Closure-chain lemma). *Let κ be an uncountable cardinal, let $1 \leq r < \omega$, and let*

$$\Phi : [\kappa]^r \longrightarrow [\kappa]^{<\omega}$$

have uniformly finite values. There is an increasing continuous chain

$$\emptyset = M_0 \subseteq M_1 \subseteq \dots \subseteq M_i \subseteq \dots \quad (i < \text{cf } \kappa)$$

such that

$$\bigcup_{i < \text{cf } \kappa} M_i = \kappa, \quad |M_i| < \kappa,$$

and every M_i is Φ -closed:

$$a \in [M_i]^r \implies \Phi(a) \subseteq M_i.$$

Proof. Choose sets $A_i \subseteq \kappa$, each of cardinality below κ , whose union is κ . At a successor stage close $M_i \cup A_i$ under Φ , iterating the operation ω times. Since Φ is finitary and finite-valued, this does not increase an infinite cardinal below κ . At a limit stage take the union. A union of an increasing chain of sets closed under a finitary operation remains closed; and because the stage is below $\text{cf } \kappa$, its cardinality remains below κ . $\square$

Write

$$I_i = M_{i+1} \setminus M_i.$$

Then $(I_i)_{i < \text{cf } \kappa}$ partitions κ , and $M_i = \bigcup_{j < i} I_j$.

2. A BIPARTITE SUBGRAPH LEMMA

The next lemma is the finite complete-bipartite consequence of Erdős and Hajnal [4, Corollary 5.6, p. 72]. We include its short closure-chain proof because the same construction is used later.

Lemma 2.1 (Uncountable chromatic number forces complete bipartite graphs). *For every positive integer n , every graph of uncountable chromatic number contains $K_{n,n}$.*

Proof. Suppose otherwise, and choose a $K_{n,n}$ -free graph G of uncountable chromatic number whose vertex cardinality κ is minimal.

For $A \in [\kappa]^n$, let

$$N(A) = \{v \in \kappa : v \text{ is adjacent to every member of } A\}.$$

Because G is $K_{n,n}$ -free, $|N(A)| < n$. Apply Lemma 1.4 to $A \mapsto N(A)$, and let M_i, I_i be the resulting chain and layers.

By minimality of κ , every induced graph $G[I_i]$ is countably chromatic. Let $G^\times$ consist of the edges joining distinct layers, and orient each such edge from the later endpoint toward the earlier endpoint. If $v \in I_i$ had n earlier neighbours, those neighbours would form an n -set $A \subseteq M_i$, and closure would give

$$v \in N(A) \subseteq M_i,$$

contrary to $v \in I_i$. Thus the orientation of $G^\times$ has outdegree at most $n - 1$. By Lemma 1.3, $G^\times$ is finitely colourable.

Colour each $G[I_i]$ with colours in ω , reusing the same palette on all layers, and combine this with a finite proper colouring of $G^\times$. The resulting countable product colouring is proper on all of G , a contradiction. $\square$

3. BIPARTITE EXPANSIONS

Rainbow bipartite submatrices.

Lemma 3.1 (Rainbow bipartite submatrices). *For all positive integers n, t , there exists $q = q(n, t)$ with the following property. Suppose the edges of $K_{q,q}$ are coloured, with an arbitrary colour set, so that at every vertex each individual colour occurs on fewer than t incident edges. Then there are n -element subsets X' and Y' of the two vertex classes such that all n^2 edges between X' and Y' have distinct colours.*

Proof. Choose independent uniformly random n -subsets X' and Y' . For every unordered pair P of distinct same-coloured edges, let I_P be the indicator that both edges of P lie in the selected $K_{n,n}$, and put

$$Z = \sum_P I_P.$$

Thus $Z = 0$ exactly when all selected edges have distinct colours.

Let m_γ be the number of edges of colour γ . The local bound gives

$$m_\gamma \leq (t - 1)q.$$

At a fixed vertex and for a fixed colour γ , fewer than t incident edges have colour γ . Summing $\binom{d_{v,\gamma}}{2} \leq (d_{v,\gamma})(t - 1)/2$ over the $2q$ vertices shows that the number of same-coloured pairs sharing a vertex is at most $(t - 1)q^2$. The total number of same-coloured pairs is at most

$$\sum_\gamma \binom{m_\gamma}{2} \leq \frac{1}{2} (\max_\gamma m_\gamma) \sum_\gamma m_\gamma \leq \frac{1}{2} (t - 1)q^3.$$

If the two edges share a vertex in the left class, their exact selection probability is

$$\frac{n}{q} \frac{(n)_2}{(q)_2};$$

the same formula holds when the shared vertex is on the right. If the two edges are disjoint, the exact probability is

$$\left(\frac{(n)_2}{(q)_2} \right)^2.$$

For $q \geq 2n$, these are at most $2n^3/q^3$ and $4n^4/q^4$, respectively. By linearity of expectation,

$$\mathbb{E}[Z] \leq \frac{2(t-1)n^3}{q} + \frac{2(t-1)n^4}{q}.$$

This is below 1 for sufficiently large q . Since Z is integer-valued and nonnegative, it cannot satisfy $Z \geq 1$ for every choice when $\mathbb{E}[Z] < 1$. Hence some choice has $Z = 0$, and the selected $K_{n,n}$ is rainbow. $\square$

The complete bipartite expansion atom.

Proposition 3.2 (The complete bipartite expansion atom). *For every positive integer n , the triple system $K_{n,n}^+$ is obligatory.*

Proof. Suppose not. Choose a $K_{n,n}^+$ -free triple system

$$H = (\kappa, E)$$

with $\chi(H) > \aleph_0$ and with κ minimal.

For a pair xy , write

$$d_H(xy) = |\{z : \{x, y, z\} \in E\}|.$$

Set $t = 3n^2 + 1$ and define a graph R on κ by

$$xy \in E(R) \iff d_H(xy) \geq t.$$

Claim 3.2.1 (the graph R is countably chromatic).

Otherwise Lemma 2.1 gives a $K_{n,n}$ in R , with core vertex set C , $|C| = 2n$. For each of its n^2 graph edges xy , choose a third vertex p_{xy} with

$$\{x, y, p_{xy}\} \in E,$$

avoiding C and all previously chosen private vertices. At every stage fewer than $2n + n^2 < t$ vertices are forbidden, while xy has at least t possible third vertices. Thus the p_{xy} can be chosen distinct and outside C , producing a copy of $K_{n,n}^+$, a contradiction. $\square$

Let

$$E_1 = \{e \in E : \text{some pair contained in } e \text{ is an edge of } R\}, \quad E_2 = E \setminus E_1.$$

A proper colouring of R is a proper colouring of (κ, E_1) , since every E_1 -edge contains an R -edge. Hence $\chi(\kappa, E_1) \leq \aleph_0$, and therefore

$$\chi(\kappa, E_2) > \aleph_0. \tag{3.1}$$

For a pair xy , define

$$\Phi(xy) = \begin{cases} \{z : \{x, y, z\} \in E\}, & d_H(xy) < t, \\ \emptyset, & d_H(xy) \geq t. \end{cases}$$

Apply Lemma 1.4 to Φ , obtaining M_i, I_i .

Every $e \in E_2$ has at least two vertices in its highest layer. Indeed, suppose the highest layer met e only in $z \in I_i$. Then the other pair $xy = e \setminus \{z\}$ lies in M_i . Since $e \in E_2$, $xy \notin R$, so $d_H(xy) < t$. Closure gives $z \in \Phi(xy) \subseteq M_i$, a contradiction.

Thus every E_2 -edge is either contained in one layer, or has exactly two vertices in its highest layer I_i and its third vertex in M_i . Let E_i^* be the E_2 -edges contained in I_i . Since $|I_i| < \kappa$ and H is $K_{n,n}^+$ -free, minimality of κ gives $\chi(I_i, E_i^*) \leq \aleph_0$. Consequently

$$E^* = \bigcup_i E_i^*$$

is countably chromatic. By (3.1), the remaining crossing-edge system

$$E^{**} = E_2 \setminus E^*$$

has uncountable chromatic number.

For each i , define a graph G_i on I_i by declaring $xy \in E(G_i)$ when there is a $\beta \in M_i$ such that $\{x, y, \beta\} \in E^{**}$. If every G_i were countably chromatic, colour each I_i properly in G_i , using the same countable palette on every layer. Every edge of E^{**} would then have differently coloured top-layer vertices, giving a countable colouring of (κ, E^{**}) , a contradiction. Hence some G_i is uncountably chromatic.

Choose $q = q(n, t)$ from Lemma 3.1. Lemma 2.1 gives a $K_{q,q}$ in G_i , with sides X, Y . For every $xy \in X \times Y$, choose

$$\beta_{xy} \in M_i$$

such that $\{x, y, \beta_{xy}\} \in E^{**}$, and regard β_{xy} as the colour of the edge xy .

This edge-colouring is locally $(t-1)$ -bounded. For example, if for fixed x and fixed β there were t different vertices y with $\beta_{xy} = \beta$, then the pair $\{x, \beta\}$ would have codegree at least t . It would belong to R , forcing the triples $\{x, y, \beta\}$ into E_1 , contrary to their membership in $E^{**} \subseteq E_2$. The same argument applies at vertices of Y .

Lemma 3.1 gives n -sets $X' \subseteq X, Y' \subseteq Y$ for which the $\beta_{xy}, x \in X', y \in Y'$, are all distinct. They lie in M_i , whereas $X' \cup Y' \subseteq I_i$, so none is a core vertex. The triples

$$\{x, y, \beta_{xy}\}, \quad x \in X', y \in Y',$$

form $K_{n,n}^+$, the final contradiction. $\square$

Corollary 3.3 (Every finite bipartite expansion is obligatory). *If J is any finite bipartite graph, then J^+ is obligatory.*

Proof. Embed the two vertex classes of J in the two classes of some $K_{n,n}$. Then J^+ is a subhypergraph of $K_{n,n}^+$. Obligatoriness is downward closed under non-induced containment: a copy of the larger finite system contains a copy of the smaller one. Apply Proposition 3.2. $\square$

Corollary 3.3 is also a consequence of Reiher [10, Theorem 1.2]; the preceding argument gives a direct proof in the present notation.

4. CLOSURE PROPERTIES

Closure under disjoint unions and one-point amalgamations is part of the established theory; see Komjáth [7, pp. 233–238] and the summary in Reiher [10, p. 1]. We include complete proofs because the rooted abundance argument is used below.

Disjoint-union closure.

Lemma 4.1 (Disjoint-union closure). *The class of finite obligatory triple systems is closed under finite disjoint unions.*

Proof. Let F_1, F_2 be obligatory and let H be uncountably chromatic. First find a copy of F_1 . Delete its finite vertex set. By Lemma 1.2 the remaining triple system is still uncountably chromatic, so it contains F_2 . The two copies are disjoint. Iteration proves the finite case. $\square$

Rooted abundance and one-point amalgamation.

Definition 4.2 (Rooted abundance). Let (F, r) be a finite rooted triple system with $|V(F)| > 1$. For a triple system H and $m \geq 1$, let $R_m(F, r; H)$ be the set of vertices v for which there are m rooted copies of F , all sending r to v , whose off-root vertex sets are pairwise disjoint.

Lemma 4.3 (Rooted-abundance lemma). *If (F, r) is finite, $|V(F)| > 1$, and F is obligatory, and if $\chi(H) > \aleph_0$, then*

$$\chi(H[R_m(F, r; H)]) > \aleph_0.$$

Proof. Put

$$B = V(H) \setminus R_m(F, r; H).$$

For each $v \in B$, consider the family of off-root vertex sets of all copies of F rooted at v . Because $v \notin R_m(F, r; H)$, this family has matching number at most $m - 1$. Choose a maximal pairwise disjoint subfamily. It contains at most $m - 1$ members; let S_v be their union. Then

$$|S_v| \leq (m - 1)(|V(F)| - 1) =: D.$$

Moreover, S_v meets the off-root part of every rooted copy at v : otherwise that off-root set would be disjoint from every member of the chosen subfamily and could be added to it, contradicting maximality.

Form a graph D_B on B by joining v to each member of $S_v \cap B$. If an edge is generated in both directions, assign it to one of its generators, and orient it away from that generator. Every vertex has outdegree at most D , so Lemma 1.3 gives a finite proper colouring of D_B .

Let A be one colour class. Then

$$S_v \cap A = \emptyset \quad (v \in A).$$

Consequently $H[A]$ contains no copy of F . Indeed, if φ were such a copy and $v = \varphi(r)$, its off-root part would have to meet S_v although both are subsets of A . Since F is obligatory, every F -free triple system is countably chromatic. Thus each of the finitely many colour classes of D_B induces a countably chromatic subsystem. Tagging a vertex by its finite D_B -colour and its countable colour inside that class gives a countable proper colouring of $H[B]$.

If $H[R_m(F, r; H)]$ were countably chromatic as well, tagging the two sides would give a countable colouring of H . This contradicts $\chi(H) > \aleph_0$. $\square$

Proposition 4.4 (One-point amalgamation closure). *The class of finite obligatory triple systems is closed under one-point amalgamations.*

Proof. Let F_1, F_2 be obligatory, with selected vertices r_1, r_2 , and let F be their one-point amalgamation. If one factor consists only of its selected vertex, the assertion is immediate. Otherwise set

$$m = |V(F_2)|.$$

By Lemma 4.3, the set

$$R = R_m(F_1, r_1; H)$$

induces an uncountably chromatic subsystem of every uncountably chromatic H . Hence $H[R]$ contains a copy of F_2 ; write φ for its embedding and let $v = \varphi(r_2)$.

At v there are m rooted copies of F_1 with pairwise disjoint off-root sets. The set $\varphi(V(F_2) \setminus \{r_2\})$ has $m - 1$ vertices, so it meets at most $m - 1$ of those off-root sets. One rooted F_1 -copy therefore avoids the F_2 -copy outside v . Their union is the required one-point amalgamation. $\square$

Corollary 3.3, Lemma 4.1, Proposition 4.4, and the trivial edgeless case prove

$$(2) \implies (1). \quad (4.1)$$

5. THE INTRINSIC DECOMPOSITION

The selected-incidence decomposition, quotient forest, and running-intersection architecture in this section follow the positive structural framework of Li [3, Sections 4–5]. We give the full argument in the present notation because its explicit assembly order is also used by the Lean development.

We prove $(2) \iff (3)$. Isolated vertices may be ignored by Lemma 1.1.

Forward preservation.

Proposition 5.1 (The intrinsic conditions are preserved by the generators and operations). *Every generator of $\mathcal{B}$ satisfies the intrinsic conditions, and the conditions are preserved under finite disjoint unions and one-point amalgamations.*

Proof. For a bipartite graph J , the expansion J^+ is linear. Every hyperedge-node is joined to its private point by a bridge. A Berge cycle cannot use a private point, so the Berge cycles of J^+ are precisely the ordinary graph cycles of J , with the same lengths; these are even. Edgeless systems satisfy the conditions vacuously, and disjoint union plainly preserves them.

Consider a one-point amalgamation. Hyperedges belonging to different factors intersect only in the identified point, so linearity is preserved. Its Levi graph is the vertex-sum of the two Levi graphs at a point-node. Every simple cycle in a vertex-sum lies wholly in one factor: a cycle crossing into the other factor would have to pass through the cut vertex twice. Consequently bridges in the factors remain bridges, and every Berge cycle lies in one factor and retains its parity. $\square$

Thus

$$(2) \implies (3). \quad (5.1)$$

Bridge-block reconstruction.

Proposition 5.2 (Bridge-block decomposition). *Suppose F is finite, has no isolated vertices, and satisfies the three intrinsic conditions. Then $F \in \mathcal{B}$.*

Proof. If $E(F) = \emptyset$, then the assumption that F has no isolated vertices gives $V(F) = \emptyset$, and F is an edgeless generator. Hence assume $E(F) \neq \emptyset$. Put $L = I(F)$, let $B(L)$ be its set of bridges, and let $\mathcal{C}$ be the set of connected components of $L - B(L)$.

Consider a hyperedge-node e . Since e is incident with a bridge and has degree three, it has at most two nonbridge incidences. It cannot have exactly one: a nonbridge incidence lies on a cycle, and that cycle must leave e through a second nonbridge incidence. Therefore every hyperedge-node has either no nonbridge incidences or exactly two.

The nontrivial components. Let $C \in \mathcal{C}$ contain a nonbridge incidence. Every hyperedge-node $e \in C$ has exactly two point-neighbours in C . Contract each such e to an ordinary graph edge between those two point-neighbours. This produces a finite graph J_C .

The graph J_C is simple: parallel edges would correspond to two hyperedges of F sharing two points, contrary to linearity. Every cycle in J_C gives a Berge cycle of the same length in F . Hence J_C has no odd cycle and is bipartite.

Each hyperedge-node $e \in C$ has a third point-neighbour p_e across a bridge. These third points satisfy:

1. $p_e \notin C$; otherwise a path in C from p_e to e , together with the incidence ep_e , would place that incidence on a cycle.
2. If $e \neq f$ lie in C , then $p_e \neq p_f$; otherwise a path in C from e to f , together with ep_e and fp_f , would place those bridge incidences on a cycle.

Define a map from J_C^+ to the subsystem P_C formed by these hyperedges as follows. Send every graph vertex of J_C to the corresponding point-node of C , send the private vertex indexed by the contracted edge associated with e to p_e , and send that expanded edge to the hyperedge e of F . The two observations above

show that the vertex map is injective: the private points are pairwise distinct and lie outside C . Every e has exactly the two contracted endpoints and p_e as its three vertices, so this map is incidence-preserving and surjective on the vertices and hyperedges of P_C . Hence it is an isomorphism $J_C^+ \cong P_C$.

Components consisting of one hyperedge-node. If a hyperedge-node e has no nonbridge incidence, it is an isolated component after the bridges are deleted. Its single triple is the expansion of a one-edge bipartite graph. Thus every hyperedge of F belongs to a piece P_C isomorphic to J^+ for a finite bipartite graph J . The hyperedge sets of these pieces partition $E(F)$. Their vertex sets cover $V(F)$: every point is incident with a hyperedge, and it either lies in that hyperedge-node's bridge-deleted component or is the point endpoint of a bridge leaving it.

Quotient forest. Contract every component of $L - B(L)$. The original bridges become edges of a quotient multigraph T . Two distinct bridges cannot join the same pair of bridge-deleted components: paths inside those two components, together with the two bridges, would form a cycle containing both, contradicting that they are bridges. Thus T is a simple graph.

To see that T is a forest, suppose that it contained a cycle. Within each contracted component on this cycle, choose a path joining the endpoints of the two incident quotient edges. Concatenating these paths with the corresponding bridges gives a closed walk in L that traverses each of those bridges exactly once. This is impossible: deleting a bridge separates its endpoints, so every closed walk crosses the resulting cut an even number of times. Hence T is a forest.

A vertex $C \in V(T) = \mathcal{C}$ is called *active* if the bridge-deleted component C contains a hyperedge-node. For an active C , the piece P_C consists of the hyperedges represented in C , all point-nodes of C , and the point endpoints of bridge incidences leaving hyperedge-nodes in C . An inactive component contains no hyperedge-node and hence no incidence of $L - B(L)$, so it is a singleton point-node.

At this stage every hyperedge already belongs to a bipartite expansion piece. The only remaining issue is global: the pieces must be ordered so that a new piece never meets the assembled system in two different points. The quotient forest records exactly the attachment geometry needed for this running intersection.

Claim (5.2.1. Running intersection). *Every component of T contains an active vertex, because F has no isolated vertices. Choose an active root in each component, and order the active vertices by nondecreasing depth, with arbitrary order among equal-depth vertices. When an active component C is added, its piece P_C meets the union of the earlier pieces in at most one point.*

Proof of the claim. For a point $p \in V(F)$, let $X(p) \in \mathcal{C}$ be the component of $L - B(L)$ containing the point-node p . The active pieces containing p are precisely $P_{X(p)}$, when $X(p)$ is active, together with the pieces P_D at active neighbours D of $X(p)$ whose corresponding original bridge has point endpoint p . Thus the vertices of T indexing pieces that contain p are contained in the closed star centred at $X(p)$.

Suppose that $p \in P_C$ also lies in an earlier piece. If $X(p) = C$, every other piece containing p is indexed by a neighbour of C . A tree has only one neighbour of C at smaller depth, namely its parent, so only that piece can be earlier. If $X(p) \neq C$, then C is a neighbour of $X(p)$. The vertex $X(p)$ cannot be a child of C , because then both $X(p)$ and every other vertex in its closed star would be deeper than C . Hence $X(p)$ is the parent of C . In either case, p is the point endpoint of the unique parent edge of C . The same reasoning applies to every point shared with an earlier piece, so all such points coincide with this parent point p_C , and

$$P_C \cap \bigcup_{D \text{ earlier}} P_D \subseteq \{p_C\}.$$

For a root active component the intersection is empty. □

Starting with the root piece in each component, add the active pieces in this order. If the new intersection is empty, use a disjoint union. If it is the singleton $\{p_C\}$, use a one-point amalgamation at that point. Several earlier pieces may already contain the same inactive attachment point, but the new step still identifies only that one existing vertex.

For completeness, the identity maps from the pieces into F assemble inductively to an incidence isomorphism from the constructed system onto the union of the pieces already added. Indeed, the running-intersection claim shows that the next piece and the preceding union have either no common vertex or exactly the one vertex used in the amalgamation; hence the construction makes no unintended

identification. The piece hyperedge sets are disjoint and partition $E(F)$, while their vertex sets cover $V(F)$. After all active pieces have been added, the resulting subsystem is therefore all of F . Different components of T are combined by disjoint union. Hence $F \in \mathcal{B}$. $\square$

This proves

$$(3) \implies (2). \quad (5.2)$$

Combining (5.1) and (5.2),

$$(2) \iff (3). \quad (5.3)$$

6. THE SEQUENCE LIFT

The one-apex construction and bridge-trace strategy were introduced by Li [3, Sections 3–4 and Theorem 4.6]. We specialise the lift to ω_1 and replace Li’s bridge-selector derivative formulation by a base-fibre and support-incidence decomposition convenient for formalisation. The detailed proof below is self-contained as an exposition, but it is not a claim of independent discovery.

The one-apex construction. Let $G = (X, A)$ be an arbitrary graph, where $A = E(G)$ is also regarded as an alphabet. Let

$$T = A^{<\omega_1}$$

be the tree of all sequences of graph edges of countable ordinal length. For $s \in T$ and $x \in X$, write $x_s = (s, x)$.

Definition 6.1 (The one-apex lift). Define the triple system $\mathcal{L}(G)$ by

$$V(\mathcal{L}(G)) = T \times X.$$

For $s \in T$, an edge $a = \{x, y\} \in A$, a sequence $t \in T$ extending $s \frown a$, and $z \in X$, put

$$\{x_s, y_s, z_t\} \in E(\mathcal{L}(G)). \quad (6.1)$$

The two vertices x_s, y_s form the base pair and z_t is the apex. Since t properly extends s , these are three distinct vertices. In the resulting triple, the sequence node occurring twice uniquely identifies s ; its two graph labels identify the edge $a = \{x, y\}$, and the remaining vertex identifies t, z . Thus each hyperedge arises from unique data, up to interchanging x and y , and the system is simple and 3-uniform.

Chromatic and finite-trace properties.

Lemma 6.2 (Chromatic lower bound). *If $\chi(G) > \aleph_0$, then*

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

Proof. Suppose $c : T \times X \rightarrow \omega$ is proper. For every $s \in T$, the restriction $x \mapsto c(x_s)$ is a countable colouring of G . Hence there is an edge

$$a_s = \{x_s^0, x_s^1\} \in E(G)$$

whose two vertices at node s have a common colour k_s .

Construct $s_\alpha \in A^\alpha$ for $\alpha < \omega_1$ by

$$s_0 = \emptyset, \quad s_{\alpha+1} = s_\alpha \frown a_{s_\alpha},$$

and at limit ordinals take unions.

If $\alpha < \beta$, then s_β extends $s_\alpha \frown a_{s_\alpha}$. If $k_{s_\alpha} = k_{s_\beta}$, take either endpoint z of the monochromatic edge at s_β . By (6.1),

$$\{(s_\alpha, x_{s_\alpha}^0), (s_\alpha, x_{s_\alpha}^1), (s_\beta, z)\}$$

is a monochromatic hyperedge. Therefore the colours k_{s_α} , $\alpha < \omega_1$, are pairwise distinct, impossible in ω . $\square$

The next result is the fibre-decomposition form of Li's bridge-trace theorem needed in this paper [3, Theorem 4.6].

Theorem 6.3 (Finite linear trace decomposition). *Every finite, not necessarily induced, linear subhypergraph K of $\mathcal{L}(G)$ is obtainable, by finite disjoint unions and one-point amalgamations, from systems J^+ where J is a finite subgraph of G .*

Proof. The following dictionary records the three local objects used in the assembly. A *base node* is a sequence s at which a lift hyperedge begins. Its two vertices with sequence coordinate s are the *core vertices*; its third vertex, whose coordinate properly extends s , is the *apex*. The *base fibre* K_s consists of all hyperedges of K based at s , together with their incident vertices. An *active base node* is one whose fibre is nonempty. The proof first identifies each fibre with one expansion atom and then shows that the fibres are attached along a forest of one-point overlaps.

First set aside every isolated vertex of K . Each is a one-vertex edgeless factor, hence isomorphic to J^+ for a one-vertex edgeless subgraph J of G , and may be restored by disjoint union at the end. We may therefore assume that every vertex of K lies in a hyperedge.

Every hyperedge of $\mathcal{L}(G)$ has a unique base node: exactly two of its vertices have sequence coordinate s , while the apex has a strictly longer sequence coordinate. We group the hyperedges of K by this coordinate and use the notation K_s from the dictionary.

Claim 6.3.1 (each base fibre is one expansion atom). For every active base node s , there is a finite subgraph J_s of G such that $K_s \cong J_s^+$.

Proof of Claim 6.3.1. For a fixed graph edge $a = \{x, y\} \in E(G)$, at most one hyperedge of K_s can have base pair $\{x_s, y_s\}$, since two such hyperedges would share two vertices and violate linearity. Let J_s be the finite subgraph of G whose edges are precisely the graph edges used by hyperedges of K_s , with vertex set consisting of their endpoints.

Define a map $J_s^+ \rightarrow K_s$ by sending each graph vertex $x \in V(J_s)$ to the core vertex x_s , and by sending the private vertex of $a = \{x, y\}$ to the apex of the unique hyperedge of K_s with base pair $\{x_s, y_s\}$. Distinct graph edges have distinct apices: if z_t were the apex for a and a' , the coordinate of t immediately after s would have to equal both a and a' . No apex is a core vertex, because an apex has sequence coordinate strictly extending s , whereas every core vertex has coordinate s . The map is therefore injective, and by construction it is bijective on hyperedges and preserves each incidence. Hence $K_s \cong J_s^+$. $\square$

Claim 6.3.2 (two fibres have at most one common point). For distinct active base nodes s and u ,

$$|V(K_s) \cap V(K_u)| \leq 1.$$

Proof of Claim 6.3.2. If a point z_t lies in both K_s and K_u , then both s and u are prefixes of t , so they are comparable. Suppose $s \subsetneq u$. A common point cannot be a core point of K_s ; it is an apex of K_s . The graph edge used by its hyperedge in K_s is the coordinate of u immediately following s . Hence all points in $K_s \cap K_u$ would be apices of the same base edge of K_s . There is at most one such apex. Therefore

$$|V(K_s) \cap V(K_u)| \leq 1. \tag{6.2}$$

$\square$

Let Q be the bipartite incidence graph of the base fibres. Its left class is the set of active base nodes. Its right class consists of points that lie in at least two fibres. Join a base node s to a point p exactly when $p \in V(K_s)$.

Claim 6.3.3 (the overlap graph is a forest). The graph Q is acyclic.

Proof of Claim 6.3.3. Suppose that Q has a cycle, and choose on it a base node s of minimum sequence length. Its two neighbouring base nodes on the cycle are proper descendants of s . The two intervening shared points are distinct, so they are distinct apices in K_s , arising from two distinct graph edges $a \neq a'$. Thus the two neighbouring base nodes lie in different immediate branches below s , one beginning with a and the other with a' .

Starting from one neighbour of s , every base node on the remaining part of the cycle lies in the same immediate branch below s . Indeed, consecutive base nodes share a point and are therefore comparable. For two comparable proper descendants of s , the first sequence coordinate after s is the same, so a length-two step through a shared point cannot move from one immediate branch to another. Induction along the remaining path therefore keeps every base node in the first branch, contradicting that its final node lies in the other branch. Hence Q is acyclic. $\square$

Root each component of Q at a base node, and order its base nodes by nondecreasing even distance from the root. For a nonroot base node s , let p_s be the point-node adjacent to s on the path to the root.

Claim (6.3.4. Base-fibre running intersection). *Every earlier factor that meets K_s meets it at p_s .*

Proof of Claim 6.3.4. Suppose K_u is earlier and contains a point $p \in K_s$. Then Q contains the length-two path $s - p - u$. Let the distance of s from the root be $2d$. The point-node p has distance either $2d - 1$ or $2d + 1$. In the second case u has distance $2d + 2$, so it cannot be earlier. Thus p has distance $2d - 1$, and the uniqueness of the path to the root in the tree Q gives $p = p_s$. This also covers an earlier sibling u , which shares the same parent point p_s . $\square$

Claim 6.3.5 (the ordered fibres assemble to K). Let U_s be the union of the factors preceding K_s . Equation (6.2) and Claim 6.3.4 give

$$V(K_s) \cap U_s \subseteq \{p_s\}.$$

Starting with the root factor, add the remaining factors in this order by disjoint unions or one-point amalgamations as appropriate. At each step the natural maps into K identify exactly the displayed common point and nothing else, so they assemble to an isomorphism onto the union of the factors already added. Different components of Q , and the isolated factors removed at the start, are combined by disjoint union. This is the running-intersection step: Claim 6.3.1 supplies the expansion atoms, while Claims 6.3.2–6.3.4 certify that each new atom is attached along at most one previous point. $\square$

Corollary 6.4 (Restrictions on finite linear traces). *Every finite linear $K \subseteq \mathcal{L}(G)$ satisfies:*

1. *every hyperedge-node of $I(K)$ is incident with a bridge;*
2. *every Berge cycle of K is contained in a single factor J_s^+ ;*
3. *a Berge cycle of length ℓ in K yields an ordinary cycle of length ℓ in G .*

Proof. In every J_s^+ the incidence with the private vertex is a bridge. Bridgehood is preserved by one-point amalgamation. A simple Levi cycle cannot cross a one-point cut, so every Berge cycle lies in one factor. Berge cycles in J_s^+ are exactly ordinary graph cycles in J_s . $\square$

7. A CLASSICAL HIGH-ODD-GIRTH INPUT

For the odd-cycle obstruction we use the following older graph theorem directly, rather than a shift-graph surrogate.

Theorem 7.1 (Erdős–Hajnal). *For every positive integer m there is a graph G such that*

$$\chi(G) > \aleph_0$$

and G contains no odd cycle of length at most m .

Source. Theorem 7.4 of Erdős and Hajnal [4, p. 76], as quoted in Erdős et al. [6, Theorem C, p. 428], gives for every positive integer i an uncountably chromatic graph containing no cycle C_{2j+1} for $0 < j < i$. Choose i so that $2i - 1 \geq m$. $\square$

8. AVOIDANCE HOSTS

The division into nonlinear, missing-bridge, and odd-cycle hosts follows the negative-half architecture of Li [3, Sections 2–4]. The first two hosts are written out in the present notation; for the third we invoke the older Erdős–Hajnal theorem from Section 7 directly.

We prove

$$\neg(3) \implies \neg(1).$$

Nonlinearity obstruction.

Proposition 8.1 (Avoidance of every nonlinear finite triple system). *There is an uncountably chromatic linear triple system. Consequently, every finite nonlinear triple system is non-obligatory.*

Proof. We use the Erdős–Rado relation [12, Theorem 4(i), formula (95), p. 471]

$$(2^{\aleph_0})^+ \longrightarrow (\aleph_1)_{\aleph_0}^2. \quad (8.1)$$

Let $\kappa = (2^{\aleph_0})^+$. Define a triple system T_κ whose vertices are the pairs in $[\kappa]^2$, and whose hyperedges are

$$\{\{\alpha, \beta\}, \{\alpha, \gamma\}, \{\beta, \gamma\}\}, \quad \alpha < \beta < \gamma < \kappa.$$

This triple system is linear: two distinct triangles of a complete graph cannot share two graph edges, because two graph edges of a triangle determine the third edge and the three underlying vertices.

It is uncountably chromatic. A countable vertex-colouring of T_κ is a countable edge-colouring of K_κ . By (8.1) there is a monochromatic subset of cardinality $\aleph_1$, and in particular a monochromatic graph triangle. This is a monochromatic hyperedge of T_κ .

If a finite triple system F is nonlinear, it has two distinct edges sharing at least two vertices. No injective embedding of those edges can exist in the linear host T_κ . Thus T_κ is F -free. $\square$

Missing-bridge obstruction.

Proposition 8.2 (Avoidance when the bridge condition fails). *Let F be finite and linear, and suppose some hyperedge-node of $I(F^\circ)$ is incident with no bridge. Then F is non-obligatory.*

Proof. This is the K_{ω_1} specialisation of Li’s sequence-lift missing-bridge obstruction [3, Section 4]. By Lemma 1.1 it is enough to avoid F° , so assume that F has no isolated vertices. Take $G = K_{\omega_1}$. Then $\chi(G) > \aleph_0$, and Lemma 6.2 gives

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

If F appeared in $\mathcal{L}(G)$, retain exactly the host hyperedges corresponding to the edges of F . Because the embedding is injective and F is linear, these selected host hyperedges form a finite linear subhypergraph isomorphic to F . Corollary 6.4 says that every hyperedge-node of such a trace is incident with a bridge, contradicting the chosen hyperedge of F . Therefore $\mathcal{L}(K_{\omega_1})$ is an uncountably chromatic F -free host. $\square$

Odd-cycle obstruction.

Proposition 8.3 (Avoidance of an odd Berge cycle). *Let F be finite and linear, and suppose F° has an odd Berge cycle. Then F is non-obligatory.*

Proof. Suppress isolated vertices and put $m = |E(F)|$. By Theorem 7.1 choose a graph G with $\chi(G) > \aleph_0$ and no odd cycle of length at most m . Lemma 6.2 gives $\chi(\mathcal{L}(G)) > \aleph_0$.

Suppose that F embeds in $\mathcal{L}(G)$ and retain exactly the host hyperedges corresponding to the edges of F . The selected trace is finite and linear. Corollary 6.4 localises every Berge cycle in one expansion fibre and sends a Berge cycle of length ℓ to an ordinary cycle of the same length in G . The given odd Berge cycle has $\ell \leq m$, contradicting the choice of G . Thus $\mathcal{L}(G)$ is an uncountably chromatic F -free host. $\square$

The three propositions cover every failure of the intrinsic conditions, and prove

$$\neg(3) \implies \neg(1). \quad (8.2)$$

9. PROOF OF THE CLASSIFICATION

Sections 3 and 4 prove $(2) \implies (1)$. Section 5 proves $(2) \iff (3)$, and Section 8 proves the contrapositive $\neg(3) \implies \neg(1)$. These implications establish all equivalences in Theorem A. In particular,

$$F \text{ is obligatory} \iff F \in \mathcal{B},$$

$$F^\circ \text{ is linear,}$$

$$F \in \mathcal{B} \iff \begin{array}{l} \text{every hyperedge-node of } I(F^\circ) \text{ is incident with a bridge,} \\ \text{every Berge cycle of } F^\circ \text{ has even length.} \end{array}$$

All embeddings are non-induced: additional host hyperedges on the image vertices are irrelevant throughout. The proof uses only ZFC, graph-colouring compactness, the classical Erdős–Hajnal high-odd-girth theorem, and the standard Erdős–Rado relation (8.1). It assumes neither CH nor GCH, and no forcing axiom or large-cardinal hypothesis.

10. FINITE CONSEQUENCES OF THE CLASSIFICATION

The classification turns several finite questions about obligatory systems into elementary questions about bipartite graphs. We record the resulting parameter spectrum and some structural and enumerative consequences. A longer proof ledger, including the source audit behind this section, is available in the accompanying repository file `FURTHER_FINITE_RESULTS.md`.

For an integer $r \geq 1$, put

$$\sigma(r) = \lceil 2\sqrt{r} \rceil.$$

For a reduced triple system F , let $c(F)$ denote the number of connected components of its Levi graph.

Proposition 10.1 (Numerical bipartite shadow). *Let F be obligatory and have no isolated vertices. If*

$$m = |E(F)|, \quad n = |V(F)|, \quad c = c(F),$$

then there is a finite bipartite graph J , without isolated vertices and with c connected components, such that

$$|E(J)| = m, \quad |V(J)| = n - m. \tag{10.1}$$

Conversely, every finite bipartite graph J without isolated vertices gives an obligatory expansion J^+ with these parameter identities and the same number of components.

Proof. Consider one connected component H of F . Proposition 5.2 and its running-intersection order write H as a succession of one-point amalgamations of connected pieces

$$J_1^+, \dots, J_k^+,$$

where each J_i is a connected finite bipartite graph. Put $m_i = |E(J_i)|$ and $v_i = |V(J_i)|$. Since a connected assembly of k pieces makes exactly $k - 1$ point identifications,

$$|E(H)| = \sum_{i=1}^k m_i, \quad |V(H)| = \sum_{i=1}^k (m_i + v_i) - (k - 1). \tag{10.2}$$

Take vertex-disjoint copies of the graphs J_i and form arbitrary one-vertex sums along a tree. Before an identification, interchange the two bipartition classes in one factor when necessary. The resulting graph J_H is connected and bipartite, and (10.2) gives

$$|E(J_H)| = |E(H)|, \quad |V(J_H)| = |V(H)| - |E(H)|.$$

The disjoint union of the J_H over the components of F is the required shadow. This shadow records the numerical data; it need not retain the attachment points of F .

Conversely, J^+ is obligatory by Corollary 3.3, and the parameter identities are immediate from the definition of expansion. $\square$

Lemma 10.2 (Connected bipartite parameters). *A connected simple bipartite graph with $r \geq 1$ edges and s vertices exists if and only if*

$$\sigma(r) \leq s \leq r + 1. \quad (10.3)$$

Equivalently,

$$s - 1 \leq r \leq \left\lfloor \frac{s^2}{4} \right\rfloor. \quad (10.4)$$

Proof. Connectedness gives $r \geq s - 1$. If the bipartition classes have sizes a, b , then

$$r \leq ab \leq \left\lfloor \frac{(a+b)^2}{4} \right\rfloor = \left\lfloor \frac{s^2}{4} \right\rfloor.$$

Conversely, the balanced complete bipartite graph $K_{\lfloor s/2 \rfloor, \lceil s/2 \rceil}$ contains a spanning tree. Starting with that tree and adding arbitrary unused edges realises every integer between $s - 1$ and $\lfloor s^2/4 \rfloor$. Finally, $r \leq \lfloor s^2/4 \rfloor$ is equivalent to $\sigma(r) \leq s$. $\square$

Theorem 10.3 (Exact finite order–size–component spectrum). *Let $m \geq 1$, $1 \leq c \leq m$, and n be integers. There exists an obligatory triple system F with no isolated vertices, exactly c connected components, and*

$$|E(F)| = m, \quad |V(F)| = n,$$

if and only if

$$\boxed{m + 2(c - 1) + \left\lceil 2\sqrt{m - c + 1} \right\rceil \leq n \leq 2m + c.} \quad (10.5)$$

Every integer in this interval occurs.

Proof. By Proposition 10.1, it is enough to determine the possible orders of a bipartite graph with m edges and c nontrivial connected components. Let the component edge counts be $m_1, \dots, m_c$, where

$$m_1 + \dots + m_c = m, \quad m_i \geq 1.$$

Lemma 10.2 shows that the i th component has between $\sigma(m_i)$ and $m_i + 1$ vertices. For $a, b \geq 1$,

$$\sigma(a) + \sigma(b) \geq \sigma(a + b - 1) + 2. \quad (10.6)$$

Indeed,

$$(\sqrt{a} + \sqrt{b} - 1)^2 - (a + b - 1) = 2(\sqrt{a} - 1)(\sqrt{b} - 1) \geq 0,$$

and taking ceilings gives (10.6). Iterating it yields

$$\sum_{i=1}^c \sigma(m_i) \geq \sigma(m - c + 1) + 2(c - 1). \quad (10.7)$$

The maximum shadow order is

$$\sum_{i=1}^c (m_i + 1) = m + c.$$

Adding the m private expansion vertices proves the two bounds in (10.5).

For attainability, take $c - 1$ shadow components equal to K_2 , and let the remaining connected bipartite component have $m - c + 1$ edges. Lemma 10.2 lets its order vary through every integer from $\sigma(m - c + 1)$ to $m - c + 2$. Expanding the resulting graph realises every integer in (10.5). $\square$

Corollary 10.4 (Connected and unrestricted spectra). *For $m \geq 1$:*

(1) *a connected obligatory reduced system with m hyperedges and n vertices exists exactly when*

$$\boxed{m + \lceil 2\sqrt{m} \rceil \leq n \leq 2m + 1}; \quad (10.8)$$

(2) *without a connectivity requirement, an obligatory reduced system with m hyperedges and n vertices exists exactly when*

$$\boxed{m + \lceil 2\sqrt{m} \rceil \leq n \leq 3m}. \quad (10.9)$$

Proof. The first statement is Theorem 10.3 with $c = 1$. For the second, let c range from 1 to m . The fixed- c intervals are consecutive, because the lower endpoint for $c + 1$ is at most one more than the upper endpoint for c by the inequality $\sigma(r) \leq r + 1$. $\square$

For a finite graph G , write $\beta(G) = |E(G)| - |V(G)| + c(G)$ for its cyclomatic number. Since

$$|E(I(F))| = 3m, \quad |V(I(F))| = n + m,$$

we have

$$\beta(I(F)) = 2m - n + c. \quad (10.10)$$

Corollary 10.5 (Exact Levi cycle-rank spectrum). *For obligatory reduced systems with $m \geq 1$ hyperedges and exactly c components, the possible Levi cycle ranks are precisely*

$$0 \leq \beta(I(F)) \leq m - c + 2 - \left\lceil 2\sqrt{m - c + 1} \right\rceil. \quad (10.11)$$

Every integer in this interval occurs. Moreover, $|V(F)| = 2m + c$ if and only if $I(F)$ is a forest; in the connected case, $|V(F)| = 2m + 1$ if and only if $I(F)$ is a tree.

Proposition 10.6 (Rigidity at two lower-endpoint families). *Let F be connected, obligatory, and reduced.*

- (1) *If $|E(F)| = t^2$ and $|V(F)| = t^2 + 2t$, then $F \cong K_{t,t}^+$.*
- (2) *If $|E(F)| = t(t + 1)$ and $|V(F)| = t(t + 1) + 2t + 1$, then $F \cong K_{t,t+1}^+$.*

Proof. At either lower endpoint, the bipartite shadow of Proposition 10.1 attains equality in the balanced bipartite edge bound. It is therefore respectively $K_{t,t}$ or $K_{t,t+1}$. For $t \geq 2$ these graphs have no cut vertex, whereas a nontrivial one-vertex sum of two or more connected bridge-block shadows has a cut vertex. Hence the canonical decomposition has one expansion piece. The cases $t = 1$ are immediate. $\square$

A forbidden-Levi formulation. Let e be a hyperedge-node with point-neighbours x, y, z . A *full theta rooted at e* consists of a vertex $w \neq e$ and three pairwise internally vertex-disjoint e – w paths whose first edges are respectively ex, ey, ez .

Lemma 10.7 (Full-theta criterion). *The node e has no incident bridge if and only if the Levi graph contains a full theta rooted at e .*

Proof. No incidence at e is a bridge exactly when x, y, z lie in one connected component after e is deleted. In that component, a minimal tree joining x, y, z has a branch point, possibly one of the terminals. Its three terminal-to-branch paths, together with ex, ey, ez , give the full theta. Conversely, a full theta supplies an alternative route around each incidence at e , so none is a bridge. $\square$

Corollary 10.8 (Forbidden Levi configurations). *A finite triple system F is obligatory if and only if $I(F^\circ)$ contains none of the following:*

- (1) *a 4-cycle;*
- (2) *a cycle of length congruent to 2 modulo 4;*
- (3) *a full theta rooted at a hyperedge-node.*

Proof. A 4-cycle in the Levi graph is exactly a pair of hyperedges sharing two points. A Berge cycle of length ℓ is a Levi cycle of length 2ℓ , so odd Berge cycles are exactly the Levi cycles of length 2 modulo 4. Lemma 10.7 translates the remaining bridge condition. Apply Theorem A. $\square$

Balanced-hypergraph consequence. A hypergraph is balanced when its incidence matrix contains no odd square submatrix having exactly two 1s in each row and column; equivalently, it has no strong odd Berge cycle. Since an obligatory system has no odd Berge cycle at all, it is balanced. The classical theorem of Berge and Las Vergnas then gives the following hereditary Kőnig property [13, 14].

Corollary 10.9 (Hereditary matching–transversal equality). *Let F be obligatory. Every hypergraph H obtained from F° by deleting vertices and/or hyperedges satisfies*

$$\nu(H) = \tau(H), \quad (10.12)$$

where ν is the maximum matching size and τ is the minimum vertex transversal size.

Colouring-polynomial factorisation. For a finite hypergraph H , let $P_H(q)$ count weak proper q -colourings and let $S_H(q)$ count strong q -colourings, in which the three vertices of each hyperedge have pairwise distinct colours. For a finite graph J and $A \subseteq E(J)$, let $k_J(A)$ be the number of connected components of the spanning graph $(V(J), A)$, including isolated graph vertices, and put

$$Q_J(q) = \sum_{A \subseteq E(J)} (-1)^{|A|} q^{|E(J)| - |A| + k_J(A)}. \quad (10.13)$$

If

$$Z_J(q, v) = \sum_{A \subseteq E(J)} q^{k_J(A)} v^{|A|}$$

is the Fortuin–Kasteleyn/Potts partition function, then

$$Q_J(q) = q^{|E(J)|} Z_J(q, -q^{-1}). \quad (10.14)$$

See White [15] for hypergraph chromatic polynomials and Sokal [16] for the multivariate Tutte/Potts framework.

Theorem 10.10 (Canonical colouring factorisation). *Let the canonical bridge-block decomposition of F° have k active pieces*

$$J_1^+, \dots, J_k^+,$$

let $c = c(F^\circ)$, let $m = |E(F)|$, and let z be the number of isolated vertices of F . If F° is nonempty, then

$$q^{k-c} P_F(q) = q^z \prod_{i=1}^k Q_{J_i}(q) \quad (10.15)$$

and

$$q^{k-c} S_F(q) = q^z (q-2)^m \prod_{i=1}^k P_{J_i}(q), \quad (10.16)$$

where P_{J_i} is the ordinary graph chromatic polynomial.

Proof. For one expansion atom, inclusion–exclusion over the events that an expanded triple is monochromatic gives

$$P_{J^+}(q) = Q_J(q).$$

Indeed, imposing the events indexed by $A \subseteq E(J)$ collapses the core vertices according to the $k_J(A)$ components of $(V(J), A)$, while the $|E(J)| - |A|$ private vertices outside A remain free. For strong colourings, the graph core must be properly coloured, and every private vertex then has $q-2$ independent choices. Thus

$$S_{J^+}(q) = (q-2)^{|E(J)|} P_J(q).$$

If H is a one-point amalgamation of H_0, H_1 , colour symmetry at the identified point gives

$$qP_H(q) = P_{H_0}(q)P_{H_1}(q), \quad qS_H(q) = S_{H_0}(q)S_{H_1}(q).$$

There are exactly $k-c$ amalgamation steps. Disjoint unions multiply the polynomials, and each isolated vertex contributes a factor q . $\square$

FORMAL VERIFICATION AND PUBLIC CHRONOLOGY

Li's v1, submitted on 23 June 2026, contains the first publicly posted complete mathematical proof of Theorem A [3, Theorem 1.1]. The public history of the present repository records a verified Lean scaffold on 15 July 2026, the complete finite constructive/intrinsic structural theorem later that day, and the complete Problem 593 endpoints by 19 July 2026. Li's v2, submitted on 23 July 2026, announced a separate Lean verification. On these public timestamps, the development here is the earliest complete public Lean formalisation located in the present audit. This statement concerns public records only and makes no inference about private development or influence. The dated commits and the scope of the comparison are recorded in `PUBLIC_CHRONOLOGY.md`.

The self-contained Lean 4 source is available in the public Erdos593 repository. For every finite triple system F , the exported theorems establish

$$\begin{aligned} \text{F.IsObligatory} &\iff \text{F.isolatedReduction.Intrinsic}, \\ \text{F.IsObligatory} &\iff \text{Constructible F.isolatedReduction}. \end{aligned}$$

Thus the development verifies both directions of the classification, including the isolated-vertex reduction. Host triple systems are not assumed finite and are quantified within the formalisation's documented ambient-universe convention. The formalisation supplements, rather than replaces, mathematical review.

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The author declares no competing interests.

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